DPCDO: Dynamic-Coordinated Learning for Multitask Domain Adaptation in Remote Sensing Imagery

Zitong Wu, Yaoyao Yin, Biao Hou, *Senior Member, IEEE*, Bo Ren, *Senior Member, IEEE*, Xianpeng Guo, and Zhengxi Guo

***Abstract*—Unified architectures that jointly perform semantic segmentation and monocular height estimation offer improved computational efficiency through shared representations, but face challenges in remote sensing applications: scarce annotations, intertask interference, and cross-regional domain shifts. These issues often lead to severe performance degradation, as models struggle with domain discrepancies and conflicting gradients. Moreover, existing domain adaptation methods remain constrained, suffering from low pseudolabel confidence, feature misalignment, and negative transfer. To address these, we propose DPCDO, a dynamic-coordinated learning approach for unsupervised multitask domain adaptation, which jointly models the multivariate conditional generation process of task outputs, domain alignment, and pseudolabel consistency. DPCDO integrates three synergistic modules: 1) dynamic patchwise prototype matching dynamically aligns local feature distributions to accommodate topographic diversity, promoting domain-invariant representation learning and cross-task information sharing; 2) multistream multiscale consistency constraint performs multiscale perturbations and enforces cross-stream consistency to enhance pseudolabel reliability; and 3) decoupled optimization for multitask heads to resolve intertask gradient conflicts. Extensive experiments show DPCDO consistently outperforms state-of-the-art methods in both semantic segmentation and monocular height estimation, validating its effectiveness in disentangling domain shifts and enhancing cross-task synergy.**

***Index Terms*—Monocular height estimation, multitask learning, semantic segmentation, unsupervised domain adaptation.**

I. Introduction

REMOTE sensing imagery plays a vital role in environmental monitoring, urban planning, and disaster management. Among various analysis tasks, semantic segmentation and height estimation are two fundamental and complementary tasks, providing semantic and geometric interpretations of the Earth's surface, respectively. Recent studies have shown that joint learning of these tasks within a unified deep network can improve representation learning efficiency, as shared features may capture richer scene-level information. However, the inherent interactions between heterogeneous tasks often lead to conflicting optimization dynamics, which can degrade performance when not properly handled [1], [2]. Notably, ground object categories such as buildings and trees exhibit strong statistical correlations with their geometric heights [3], [4], indicating that semantic and geometric cues are intrinsically coupled. This suggests that semantic segmentation and height estimation are well-suited for joint learning, provided that the network is structurally designed to explicitly model task relationships, enabling cross-task feature synergy while mitigating gradient interference. Therefore, how to fully exploit the intrinsic correlation between semantic segmentation and height estimation while effectively mitigating task conflicts in multitask joint learning has become a critical and significant issue in multitask remote sensing learning.

Obtaining dense and high-fidelity annotations for remote sensing imagery is economically expensive and labor intensive [5], [6], [7], [8], which largely limits the applicability of supervised learning methods in large-scale remote sensing scenarios and makes unsupervised learning a viable alternative.

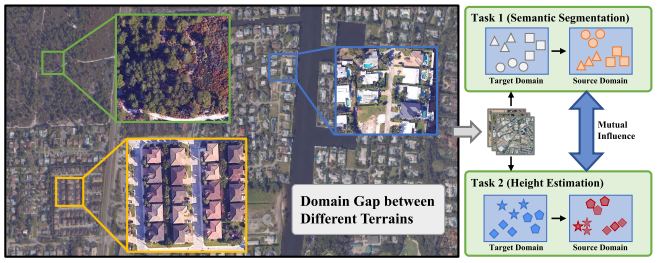


Fig. 1. Remote sensing image illustrating significant domain gaps across diverse terrains (e.g., forests, urban areas, and buildings) with heterogeneous spectral and geometric patterns. The visual discrepancies highlight the challenge of cross-task interference during joint training of semantic segmentation and height estimation, where conflicting feature representations and gradient updates destabilize model convergence in multitask learning.

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The authors are with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education of China, and the Joint International Research Laboratory of Intelligent Perception and Computation, Xidian University, Xi'an 710071, China (e-mail: wuzitong@xidian.edu.cn; 1914423532@qq.com; avcodec@hotmail.com; asdwer2046@126.com; guoxianpeng@xidian.edu.cn; XDU_Gzx@stu.xidian.edu.cn).

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However, significant multidomain discrepancies arising from differences in imaging parameters, sensor-specific artifacts, illumination variations, and heterogeneous geomorphological patterns fundamentally undermine the cross-domain adaptation capability of multitask learning architectures [9], [10], [11], [12], [13], [14], as shown in Fig. 1. Unsupervised domain adaptation (UDA) enables knowledge transfer from labeled source domains to unlabeled target domains, thereby substantially reducing dependence on high-quality annotations while improving cross-domain adaptation capability by narrowing the distribution gap between the source and target domains [15], [16], [17], [18], [19], [20]. Existing UDA approaches exhibit significant limitations: negative transfer mitigation methods are often confined to coarse-grained global feature alignment [21], [22], failing to capture local heterogeneity in remote sensing imagery; pseudolabel generation methods typically rely on single-stream or single-scale constraints [23], [24], [25], [26], [27], inadequately characterizing complex feature variations; multitask learning frameworks face dual limitations: insufficient cross-task information sharing and reliance on complex gradient interference suppression mechanisms [1], [28], [29], which increase model complexity and hinder the generalizability and portability of the approach. Therefore, how to effectively model domain shifts in a multitask setting and achieve reasonable synergy among the optimization processes of different tasks remains a problem worthy of in-depth research.

Building upon the RS3DAda [30], we introduce DPCDO, a method designed to systematically address the intertwined challenges of domain shift and task conflict. DPCDO integrates spatial selectivity, prediction consistency modeling, and task coupling into a unified framework for multidomain adaptation in remote sensing. This method effectively models the inherent structural correlations between tasks, such as confidence supervision, and captures the differential impacts of domain shifts on the model by introducing probabilistic dependencies between modules. Through dynamic regulation mechanisms, it adaptively modulates the intensity of pixel-level supervisory signals and gradient optimization pathways, based on variations in image complexity, task consistency, and domain discrepancies. Extensive experiments on multiple benchmark datasets demonstrate that DPCDO outperforms the state-of-the-art multitask models in both height estimation and semantic segmentation for unsupervised domain adaptation tasks. Notably, DPCDO introduces only a minimal number of additional learnable parameters, is easy to integrate, and exhibits strong domain adaptability. As a result, it offers an efficient, universal solution for remote sensing multitask learning in computationally constrained environments. The code is publicly available at: https://github.com/yyy-123-abc/DPCDO. The key contributions of our work are summarized as follows.

1) We propose the DPCDO method, which systematically addresses the challenges of domain shift and task conflict in multitask unsupervised domain adaptation. Experimental results on multiple benchmark remote sensing datasets demonstrate that DPCDO achieves state-of-the-art performance in both semantic segmentation and height estimation tasks within the unsupervised domain adaptation framework.
2) We introduce the dynamic patchwise prototype matching (DPPM) module, which enables spatial alignment of local features between the source and target domains to mitigate negative transfer. In addition, it leverages cross-task domain adaptation weights to facilitate the joint optimization of semantic segmentation and height estimation.
3) We introduce a multistream multiscale consistency constraint (MS2C), which enhances cross-stream and cross-scale consistency under various perturbations and strict thresholds, while progressively adjusting the intensity of pseudosupervision.
4) We introduce decoupled optimization for multitask heads (DOMHs), a lightweight strategy that alternately updates task-specific decoders to effectively mitigate gradient interference between tasks while preserving multitask collaboration through a shared encoder.

II. Related Work

A. Monocular Height Estimation

Monocular height estimation is a critical task in remote sensing, aiming to predict height values from a single image. This task faces significant challenges due to the absence of geometric cues typically available in stereo or multiview imagery. Inspired by advances in monocular depth estimation, deep neural networks have been widely adopted for monocular height estimation [31], [32], [33]. Some approaches employ encoder–decoder architectures to regress height values [34], while others utilize generative adversarial networks to synthesize height maps [35]. Semantic labels are frequently incorporated as effective priors, with several studies proposing multitask networks for joint building footprint extraction and height estimation [31], [3], [36]. Other methods leverage semantic labels as auxiliary information to enhance height estimation [37].

However, most existing methods rely heavily on full supervision, requiring large amounts of annotated data that are costly and time consuming to acquire. To address this limitation, UDA methods have gained increasing attention. These approaches transfer knowledge from labeled source domains to unlabeled target domains, thereby reducing dependence on expensive annotations. Despite their potential, the application of UDA to monocular height estimation remains relatively unexplored, with only preliminary studies available [38], [30]. Consequently, developing effective UDA frameworks for this domain represents an important direction for future research.

B. Self-Training Approaches and Pseudolabel Quality in UDA

UDA aims to bridge domain gaps by transferring knowledge from label-rich source domains to unlabeled target domains. Due to the high cost and difficulty in obtaining labeled remote sensing images, UDA has become an effective solution to address this issue [39], [40], [41], [42]. Self-training approaches, particularly the teacher–student framework, have

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emerged as the dominant UDA paradigm. In this framework, teacher models generate pseudolabels for target domain data, and student models use these pseudolabels for training [23], [26], [30], [43], [44], [45], [46], [47]. The quality of pseudolabels is crucial for model performance, especially in complex and highly heterogeneous fields such as remote sensing [48], [49], [30]. Pseudolabel generation methods typically rely on confidence thresholding [23], [24], [50] and consistency regularization strategies [25], [26], [27]. Confidence thresholding filters high-quality pseudolabels by setting a confidence threshold, while consistency regularization methods improve the reliability of pseudolabels by applying multisample or different perturbations. Although these methods improve pseudolabel quality to some extent, most consistency methods [25], [26], [27] still depend on single-stream or single-scale supervision mechanisms. This makes them insufficient to comprehensively capture the complex feature variations in remote sensing imagery under diverse imaging conditions and significant domain discrepancies.

C. Prototype-Based Feature Alignment and Negative Transfer Mitigation

Negative transfer, where a subset of source-domain samples adversely affects domain adaptation, is a key challenge in UDA [17], [51]. In remote sensing imagery, this problem is further exacerbated by substantial discrepancies between source and target domains in terms of terrain, climate conditions, and sensor types, which makes effective knowledge transfer particularly difficult. Consequently, mitigating negative transfer is crucial for improving both the performance and the generalization ability of remote sensing image analysis models.

Existing selective adaptation strategies alleviate negative transfer by assigning weights to source-domain samples [21], [22], [52]. However, most of these methods rely on global weighting mechanisms, where a single weight is computed for each image to guide source-sample selection. This design suffers from two limitations. On the one hand, images with low overall weights may still contain local regions that are highly beneficial for the target domain, which are thus underutilized. On the other hand, even in high-weight images, there may exist local regions that are detrimental to transfer, yet they are insufficiently suppressed. As a result, global weighting strategies fail to finely distinguish and handle locally beneficial or harmful information, limiting their effectiveness in remote sensing image analysis. In recent years, prototype-based methods have attracted increasing attention due to their advantages in capturing semantic features [53], [54], [55], [56], [57], [58]. Nevertheless, their potential in multitask frameworks for jointly leveraging semantic and height information remains underexplored.

To address this gap, we propose a prototype-based fine-grained domain alignment mechanism that generates domain adaptation weight maps to mitigate negative transfer, promote cross-task information sharing, and enable the collaborative optimization of semantic segmentation and height estimation.

D. Multitask Learning

Multitask learning (MTL) enhances model performance by exploiting structured relationships and complementary information across tasks [59], [60], [61], [62]. In remote sensing applications, tasks such as semantic segmentation, height estimation, and change detection are inherently correlated, sharing similar spatial structures, spectral characteristics, and land-cover patterns. This makes MTL particularly well-suited for such settings. However, differences in task objectives, conflicting gradient directions, and varying task difficulties often give rise to gradient interference, which destabilizes joint optimization and degrades overall performance [1], [2].

To mitigate gradient conflicts, a number of gradient-balancing and task-weighting strategies have been proposed, including GradNorm for dynamically adjusting task loss weights [1], multiobjective optimization based on Pareto optimality [2], and uncertainty-based task weighting [29]. While effective to an extent, these methods generally rely on complex optimization procedures, additional gradient projection operations, or substantial architectural modifications. Such overhead limits their scalability in remote sensing scenarios, where multitask models typically need to process large-scale, high-resolution data and may operate under constrained computational resources (e.g., onboard or edge deployments). In addition, existing domain adaptation approaches have not fully leveraged the shared feature representations inherent in MTL, as they often lack mechanisms for coordinated cross-task domain alignment. This results in suboptimal transfer of source-domain knowledge and ultimately weakens model robustness and adaptation capability in cross-domain remote sensing settings.

To address these challenges, we introduce a dynamic-coordinated learning framework that performs lightweight decoupled optimization and fine-grained cross-task information sharing. Our approach alleviates gradient conflicts while dynamically modeling intertask dependencies, enabling effective and scalable multitask domain adaptation without imposing substantial computational or architectural burden.

III. Preliminaries

Semantic segmentation and height estimation represent two fundamentally different computational paradigms—discrete classification and continuous regression, respectively. Despite this distinction, the two tasks exhibit strong statistical correlations in structured environments such as remote sensing imagery. A deep understanding of their mathematical differences and potential connections is crucial for designing efficient multitask architectures and domain adaptation strategies capable of handling both tasks jointly. This section provides an in-depth analysis of their theoretical foundations, highlights key differences and potential connections, and discusses the main challenges involved in integrating them within a unified multitask learning framework.

A. Task Discrepancy in Joint Learning

1) Semantic Segmentation: Pixelwise Classification: Semantic segmentation is a typical dense classification problem, aiming to assign a predefined semantic category—such

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TABLE I
MATHEMATICAL COMPARISON OF OUTPUT SPACES BETWEEN SEMANTIC SEGMENTATION AND HEIGHT ESTIMATION

Attribute	Segmentation	Height Estimation
Output Dimension	$\mathbb{R}^C$ (probability vector)	$\mathbb{R}_{\geq 0}$ (scalar value)
Constraint	$\sum_c \hat{Y}_{\text{seg}}^{i,c} = 1$	No constraint
Decision Rule	$\arg\max_c \hat{Y}_{\text{seg}}^{i,c}$	$\hat{Y}_{\text{height}}^{i}$

as buildings, vegetation, water bodies, or ground—to each pixel. This task maps continuous spectral and spatial features to discrete class labels through a probabilistic distribution. Mathematically, it can be formulated as a mapping via a probability distribution

$$\hat{Y}_{\text{seg}}^{i} = \text{Softmax}\left(W_{\text{seg}} \cdot f_\theta(x_i) + b_{\text{seg}}\right) \in \mathbb{R}^C \tag{1}$$

where $\hat{Y}_{\text{seg}}^{i}$ represents the probability distribution over C classes for pixel i, satisfying $\sum_{c=1}^{C} \hat{Y}_{\text{seg}}^{i,c} = 1$. The function $f_\theta(x_i)$ denotes the feature extractor, while W_{seg} and b_{seg} are the parameters of the classification layer.

2) Height Estimation: Pixelwise Regression: Height estimation is a typical dense regression problem, aiming to predict a continuous elevation value for each pixel. This task maps input features to specific values in a continuous numerical space through regression.

Mathematically, it can be formulated as a scalar regression

$$\hat{Y}_{\text{height}}^{i} = W_{\text{height}} \cdot f_\theta(x_i) + b_{\text{height}} \in \mathbb{R}_{\geq 0} \tag{2}$$

where $\hat{Y}_{\text{height}}^{i}$ denotes the predicted height value for pixel i (a nonnegative real number). W_{height} and b_{height} are the parameters of the regression layer, without any probabilistic normalization constraints.

3) Mathematical Output Space Comparison: Table I presents the mathematical comparison of the output spaces between semantic segmentation and height estimation.

4) Mathematical Formulation of Joint Modeling: When the two tasks are jointly trained through a shared backbone network, the model must simultaneously satisfy both classification and regression constraints and requirements

$$\begin{cases} \text{Shared features: } f_\theta(x) & = \text{Backbone}(x) \\ \text{Segmentation head: } \hat{Y}_{\text{seg}} & = \text{Softmax}\left(W_{\text{seg}} f_\theta(x) + b_{\text{seg}}\right) \\ \text{Height head: } \hat{Y}_{\text{height}} & = W_{\text{height}} f_\theta(x) + b_{\text{height}}. \end{cases} \tag{3}$$

Classification tasks typically require the model to learn features with category separability, maximizing the distinction between different classes. On the contrary, regression tasks emphasize numerical continuity, encouraging similar feature representations for pixels with similar target values. This fundamental difference in feature learning objectives represents a key challenge that must be carefully balanced in joint modeling.

B. Conditional Probabilistic Task Modeling

Semantic categories and height values exhibit strong statistical correlations in real-world remote sensing data. Buildings typically have higher elevations than roads, trees show characteristic height distributions, and water bodies maintain consistent elevation levels.

The statistical relationship between semantic categories and height distributions can be explicitly modeled using conditional probability

$$P\left(\hat{Y}_{\text{height}} \mid \hat{Y}_{\text{seg}} = c\right) \sim \mathcal{N}\left(\mu_c, \sigma_c^2\right) \tag{4}$$

where μ_c and σ_c^2 denote the mean and variance of the height values corresponding to class c. In joint learning, the model can capture this latent relationship through the following formulation:

$$\mathcal{L}_{\text{joint}} = \mathcal{L}_{\text{seg}} + \alpha \sum_{c=1}^{C} \left\| \hat{Y}_{\text{height}} - \mu_c \right\|^2 \cdot \mathbb{I}\left(\hat{Y}_{\text{seg}} = c\right) \tag{5}$$

where α is a weighting coefficient, and $\mathbb{I}(\cdot)$ is an indicator function that ensures the predicted height aligns with the expected height distribution μ_c of the corresponding class. This formulation allows the model to learn class-conditional height priors, thereby improving the efficiency of feature learning.

C. Challenges and Implications for Joint Modeling

The fundamental differences between semantic segmentation and height estimation can be attributed to their distinct mathematical natures. Semantic segmentation is a classification task that produces a discrete probability distribution, with its decision process driven by probabilistic competition. Optimization focuses on forming clear class boundaries and a sharp separation in the feature space. On the contrary, height estimation is a regression task that outputs continuous scalar values, with its decision process being a direct mapping. Optimization emphasizes accuracy and the preservation of spatial continuity in the predicted values. These differences in task characteristics pose significant challenges for joint modeling. In particular, the segmentation branch favors discrete, clustered features, while the regression branch requires smooth, continuous mappings—creating inherent conflicts in shared feature representations. Balancing these opposing demands necessitates careful network design, such as using task-specific decoupling mechanisms or conditional priors.

IV. METHODOLOGY

A. Overview

We propose DPCDO, a dynamic-coordinated learning approach for multitask UDA in remote sensing imagery, aiming to jointly mitigate domain shifts and intertask conflicts through unified dynamic-coordinated learning, as shown in Fig. 2. Operating under a teacher–student paradigm [63], DPCDO leverages labeled source data (X_s with y_{Seg}^{s}, y_H^s) and unlabeled target images (X_t) to jointly perform semantic segmentation and monocular height estimation across domains. DPCDO consists of three key modules: DPPM, MS2C, and DOMH, with the core idea of learning coordinated conditional distributions. Since the three modules share feature extractors,

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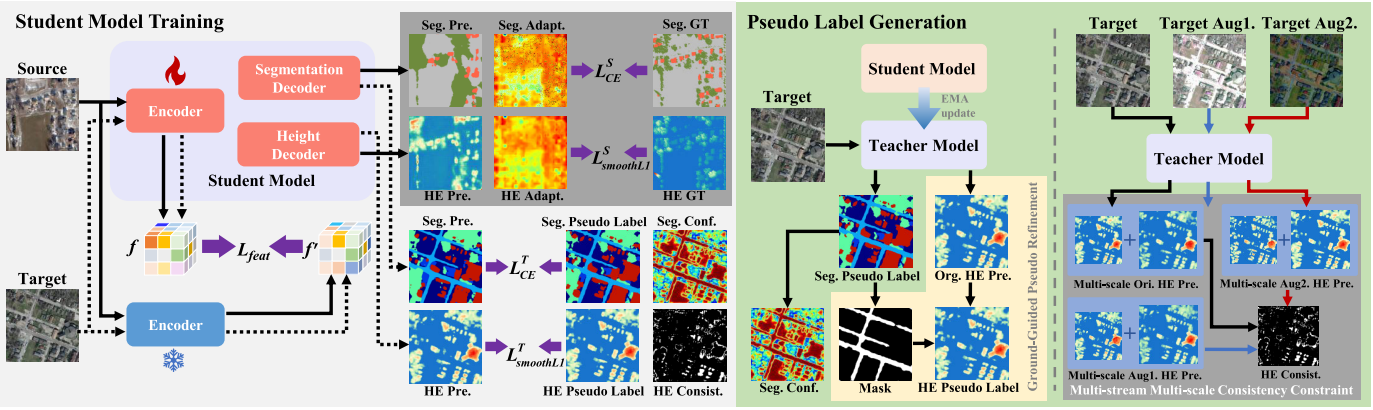


Fig. 2. Overview of the proposed DPCDO: the framework adopts a teacher–student network architecture, where the teacher model generates pseudolabels for the target domain, and the student model undergoes joint training by integrating source-domain supervision, target domain pseudolabels, and feature constraints, with the teacher's parameters updated via the EMA mechanism. Key modules (gray regions) include: the DPPM module in the student network that generates domain-adaptive weight maps to suppress negative transfer effects and facilitate cross-task knowledge sharing; the MS2C module in the teacher network that enhances pseudolabel quality through multiscale multiaugmentation consistency constraints.

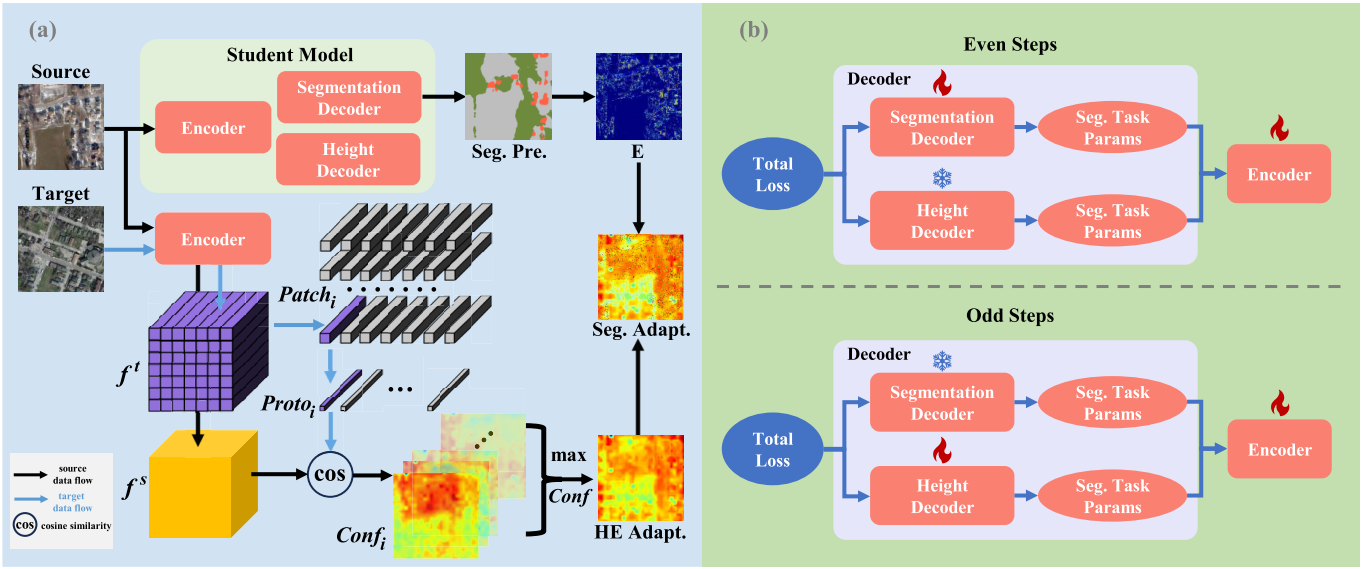


Fig. 3. (a) Process of generating domain adaptation weight maps via DPPM. Black solid line: source image. Blue solid line: target image. (b) Gradient backpropagation process of DOMH.

strict conditional independence assumptions do not hold, and we employ a structured conditional mixture approximation

$$P(Y \mid X, D) \approx \sum_{m \in M} P(Y \mid X, D, m) \cdot P(m \mid X, D) \tag{6}$$

where $Y = \{Y_{\text{seg}}, Y_h\}$ represents multitask outputs, $X = \{X_s, X_t\}$ represents source and target domain input images, $D \in \{s, t\}$ denotes domain labels and $m \in M = \{\text{DPPM}, \text{MS2C}, \text{DOMH}\}$ are module indices. The three modules are functionally complementary: DPPM addresses spatial transfer, MS2C handles pseudosupervised learning, and DOMH controls gradient conflicts while performing probabilistic module fusion. The dynamic-coordinated learning adaptively balances multiple optimization objectives, thereby enhancing robustness and training stability in cross-domain multitask learning.

B. Dynamic Patchwise Prototype Matching

In DPCDO's dynamic-coordinated learning approach, DPPM provides localized domain alignment conditional transfer pathways $P_{\text{DPPM}}(Y \mid X, D)$, offering "spatial selectivity" aligned representations in domain adaptation to mitigate cross-domain local representation shift. Unlike traditional methods that typically directly discard low-quality source images, leading to the loss of useful pixel information, DPPM constructs fine-grained domain adaptation weight distribution maps by extracting patchwise prototypical representations from deep-level feature representations of the shared encoder, as illustrated in Fig. 3(a). The weight distribution map can precisely identify and highlight pixel-level regions conducive to cross-domain knowledge transfer, effectively suppressing negative transfer effects through adaptive weight allocation mechanisms. Notably, this domain adaptation weight map is simultaneously applied to both height estimation and semantic

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segmentation tasks, enabling cross-task collaborative optimization and feature sharing. Meanwhile, we introduce a random source-target pairing mechanism that effectively prevents overfitting to specific domains. This method achieves robust domain adaptation without requiring domain-specific supervision while maintaining compatibility with standard UDA pipelines.

1) Generation of Domain Adaptation Weight Map: At each iteration, we randomly sample a target domain image. Unlike previous works that partition patches on segmentation decoder features, we extract and reshape target domain image features from the fourth layer of the encoder $f^t \in R^{C_f \times H_f \times W_f}$ into patches $p^t \in R^{n \times C_f \times P^2}$. Encoder features exhibit stronger task-agnostic properties and generalization capabilities, providing shared semantic representations for multiple dense prediction tasks, which enables our prototype matching approach to effectively extend to other tasks, such as height estimation. Here, H_f, W_f, and C_f represent the height, width, and channel of f^t, respectively. The patch size is P with a total of n patches, and there is no overlap between patches. The prototype for each patch is computed as follows:

$$proto_i = \frac{1}{P^2} \sum_{u=1}^{P} \sum_{v=1}^{P} patch_i(u, v) \tag{7}$$

where $proto_i \in \mathrm{R}^{C_4}$, $i \in [0, n-1]$ and (u, v) specifies each position in the patch. We compute the similarity between each prototype and the source features as a confidence map $Conf_i \in \mathrm{R}^{C_4 \times H_4 \times W_4}$ for adaptation by

$$Conf_i = F(f^s, proto_i) \tag{8}$$

where $f_4^s \in \mathrm{R}^{C_4 \times H_4 \times W_4}$ is the source-image feature obtained from *layer4*, and we choose the cosine similarity as the distance function $\mathcal{F}$. We then perform a max operation across all prototypes to obtain the $Conf \in \mathrm{R}^{H_4 \times W_4}$ for this source sample in this running iteration by

$$Conf = \max_{i \in [0, n-1]} Conf_i. \tag{9}$$

The patchwise prototype approach for confidence map computation offers two advantages: 1) higher computational efficiency compared to pixelwise calculations and 2) patch-level aggregation effectively suppresses pixel-level noise interference caused by domain gaps. Then, $Conf$ is upsampled to the original image size and used directly as the domain adaptation weight map for height estimation, denoted as C_H^{s}. For the domain adaptation weight map in semantic segmentation, we rectify the confidence map based on the entropy of the source prediction. Given the source prediction pX_s' after statistical image transformation [simple histogram matching (HM) and pixel distribution matching (PDM)], we calculate its entropy map $E \in \mathrm{R}^{H_4 \times W_4}$ as follows:

$$E = -\frac{1}{\log(C)} \sum_{c=1}^{C} p_{X_s'}^c \log\left(p_{X_s'}^c\right) \tag{10}$$

where C denotes the number of classes, $p_{X_s'}^c$ is defined as the predicted probability of X_s' for class c. Then, E is upsampled to the original image size, and the rectified confidence map is achieved by

$$C_{Seg}^{\mathrm{s}} = Conf \cdot (1 - E). \tag{11}$$

High entropy indicates low confidence for the prediction; therefore, $(1 - E)$ can highlight the confident region based on the prediction. The thought behind this design is that the source should be confident enough to help the adaptation to the target.

2) Source-Domain Loss: The source-domain supervision loss L_{source} is computed using the domain adaptation weight maps C_{Seg}^{s} and C_H^{s} as follows:

$$L_{source} = \frac{1}{N} \sum_{j=1}^{N} \Big(L_{CE}\left(\hat{Y}_{Seg}^{s(j)}, y_{Seg}^{s(j)}\right) \cdot C_{Seg}^{s(j)} + L_{SmoothL1}\left(\hat{Y}_H^{s(j)}, y_H^{s(j)}\right) \cdot C_H^{s(j)} \Big) \tag{12}$$

where $\hat{Y}_{Seg}^{s(j)}$ and $\hat{Y}_H^{s(j)}$ denote the semantic segmentation and height estimation predictions of X_s', obtained from the student network, respectively. L_{CE} denotes the cross-entropy loss, $L_{SmoothL1}$ denotes the smooth $L1$ loss, and N is the total number of samples.

C. Multistream Multiscale Consistency Constraint

In DPCDO's dynamic-coordinated learning approach, MS2C provides pseudosupervised consistency enhancement pathways $P_{\mathrm{MS2C}}(Y \mid X, D)$ by modeling pseudolabel reliability, motivated by two domain-specific challenges in remote sensing imagery. First, remote sensing data acquired under varying conditions is susceptible to sensor noise, atmospheric interference, and illumination variations, resulting in inconsistent image quality. To address this, MS2C incorporates remote sensing-tailored augmentations and enforces cross-stream consistency constraints, thereby improving model robustness and generalization under suboptimal imaging conditions. Second, near-nadir remote sensing imagery provides limited geometric cues for height inference, making monocular height estimation prone to systematic bias. To mitigate this, MS2C introduces multiscale consistency constraints across decoder layers and applies strict thresholding to filter reliable height pseudolabels, reducing overreliance on predictions from the final layer alone.

1) Consistency Constraint Formulas: The MS2C is introduced in the teacher model to filter out more reliable pseudolabels through a generated height consistency map. For target domain images, MS2C employs several remote sensing-tailored augmentation strategies.

1) Color jittering, which adjusts brightness, contrast, saturation, and hue to simulate illumination variations and sensor response differences;
2) blur perturbation, which applies a Gaussian blur to simulate resolution degradation from atmospheric scattering or altitude changes;
3) noise perturbation, which injects zero-mean Gaussian noise with moderate variance to emulate common sensor noise.

During training, these operations are dynamically sampled to generate diverse perturbation views, capturing appearance

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variations under different imaging conditions. Predictions are then made on feature maps at multiple scales, enforcing height map consistency across both augmentation streams and scales. The multistream consistency constraint is defined as

$$C_{H1}^{t} = I\left(\begin{array}{c}\max\left(\frac{\tilde{Y}_H^{t,aug1}}{\tilde{Y}_H^{t,ori}}, \frac{\tilde{Y}_H^{t,ori}}{\tilde{Y}_H^{t,aug1}}\right) \le \eta_1 \\ \wedge \max\left(\frac{\tilde{Y}_H^{t,aug2}}{\tilde{Y}_H^{t,ori}}, \frac{\tilde{Y}_H^{t,ori}}{\tilde{Y}_H^{t,aug2}}\right) \le \eta_1\end{array}\right) \tag{13}$$

where $\tilde{Y}_H^{t,aug1}$ and $\tilde{Y}_H^{t,aug2}$ denote the height estimation predictions of the target domain image after different augmentations, while $\tilde{Y}_H^{t,ori}$ denotes the original height estimation prediction. The map C_{H1}^{t} is a height consistency map generated by thresholding with η_1, where I is the indicator function. The same procedure is applied to the height estimation results predicted from another layer in the decoder, which serves as an auxiliary layer to provide additional supervision, as shown below

$$C_{H2}^{t} = \left(\begin{array}{c}\max\left(\frac{\tilde{Y}_H^{t,aux_aug1}}{\tilde{Y}_H^{t,aux_ori}}, \frac{\tilde{Y}_H^{t,aux_ori}}{\tilde{Y}_H^{t,aux_aug1}}\right) \le \eta_2 \\ \wedge \max\left(\frac{\tilde{Y}_H^{t,aux_aug2}}{\tilde{Y}_H^{t,aux_ori}}, \frac{\tilde{Y}_H^{t,aux_ori}}{\tilde{Y}_H^{t,aux_aug2}}\right) \le \eta_2\end{array}\right) \tag{14}$$

where C_{H2}^{t} is a height consistency map generated based on the threshold η_2. The final height consistency map C_H^t is obtained by combining C_{H1}^{t} and C_{H2}^{t} as follows:

$$C_H^t = C_{H1}^{t} \wedge C_{H2}^{t}. \tag{15}$$

A pixel is regarded as reliable only if the prediction results from multiple augmentation streams remain consistent with the original image at each scale.

2) Target Domain Loss: Height pseudolabels are refined using a ground mask $\mathcal{G}$, generated from the semantic segmentation pseudolabels $\tilde{Y}_{Seg}^{t}$ as $\mathcal{G} = I(\tilde{Y}_{Seg}^{t} = Ground)$. The correction process is defined as

$$\tilde{Y}_H^{t(j),refined} = \tilde{Y}_H^{t(j),ori} \cdot (1 - \mathcal{G}). \tag{16}$$

This strategy prevents the use of incorrect ground height data during training and facilitates more accurate identification and learning of height features from target categories such as trees and buildings. The target domain loss function is formulated as

$$L_{target} = \frac{1}{N}\sum_{j=1}^{N}\left(L_{CE}\left(\hat{Y}_{Seg}^{t(j)}, \tilde{Y}_{Seg}^{t(j)}\right) \cdot C_{Seg}^{t(j)} + L_{SmoothL1}\left(\hat{Y}_H^{t(j)}, \tilde{Y}_H^{t(j),refined}\right) \cdot C_H^{t(j)}\right) \tag{17}$$

where $\hat{Y}_{Seg}^{t(j)}$ and $\hat{Y}_H^{t(j)}$ denote the semantic segmentation and height estimation predictions produced by the student network for the target domain image, respectively. The confidence map $C_{Seg}^{t(j)} = I(\tilde{Y}_{Seg}^{t} > \tau)$ is computed by applying a threshold τ to the segmentation probabilities of the pseudolabels. In the multitask unsupervised domain adaptation setting, errors in pseudolabels can be repeatedly reinforced within a single task and further propagated across tasks through shared representations. To mitigate the negative impact of noisy semantic pseudolabels on joint optimization, we adopt a relatively high confidence threshold $\tau = 0.95$, retaining only highly confident pixel predictions as supervision signals, thereby improving the stability and robustness of the training process.

D. Decoupled Optimization for Multitask Heads

In DPCDO's dynamic-coordinated learning approach, DOMH provides optimization pathway switching strategies $P_{\mathrm{DOMH}}(Y \mid X, D)$ between multitasks, controlling task gradient update pathways and offering an "optimization conflict avoidance mechanism" in multitask collaboration. In remote sensing imagery, semantic categories and geometric heights exhibit strong statistical correlations—buildings typically have higher elevations than roads, trees show characteristic height distributions, and water bodies maintain consistent levels. This inherent coupling motivates employing a shared encoder to jointly learn complementary semantic-geometric features. However, in MTL, the "synchronous" optimization of heterogeneous tasks—discrete classification for segmentation versus continuous regression for height estimation—often suffers from gradient interference issues. The segmentation and regression heads may constrain each other, potentially leading to performance degradation in one of the tasks. DOMH, as a decoupled multitask head optimization strategy, can decouple the gradient updates of the segmentation and regression heads while preserving the model's multitask collaboration capability. An alternating optimization strategy is adopted for semantic segmentation and height estimation tasks.

1) *Odd Steps:* The Seg decoder and encoder are updated while the HE decoder parameters are frozen.
2) *Even Steps:* The HE decoder and encoder are updated while the seg decoder parameters are frozen.

The gradient backpropagation process of DOMH is illustrated in Fig. 3(b).

E. Overall Loss

To mitigate overfitting between the source and target domains, a frozen DINOv2 [64] encoder is utilized to guide the updates of the student encoder. Let f denote the features of the student encoder, and f' denote the features of the frozen DINOv2 encoder. A cosine similarity constraint is imposed on the feature updates, with the loss function defined as follows:

$$\mathcal{L}_{feat} = \begin{cases}1 - \frac{f \cdot f'}{\|f\|\|f'\|}, & \text{if } \frac{f \cdot f'}{\|f\|\|f'\|} < \epsilon \\ 0, & \text{otherwise}\end{cases} \tag{18}$$

where ϵ is the threshold. The overall loss of the model is formulated as

$$L_{overall} = L_{source} + \lambda_{target} L_{target} + \lambda_{feat} L_{feat}. \tag{19}$$

V. Experiments

A. Datasets

1) Source-Domain Dataset: We use SynRS3D [30] as the source domain. This dataset is a synthetic dataset for global 3-D semantic understanding of monocular remote sensing imagery, with a ground sampling distance (GSD) range from 0.05 to 1 m. SynRS3D [30] consists of a total of 69,667 optical images with a resolution of 512 × 512, covering six different urban styles and encompassing eight land cover types, precise height information, and building change masks. The dataset provides comprehensive geographic coverage and detailed annotation information, offering strong support for various tasks in the field of remote sensing.

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TABLE II
INTRODUCTION TO TARGET DOMAIN DATASETS

Domain	Datasets	Region	Height mean&std	Channel
T.D.1	Houston [65]	US	[3.07, 5.02]	RGB
	JAX [66]	US	[4.73, 9.02]	RGB
	OMA [66]	US	[2.37, 5.27]	RGB
	GeoNRW_Urban [67]	Germany	[2.46, 4.31]	RGB
	GeoNRW_Rural [67]	Germany	[2.03, 4.21]	RGB
T.D.2	ATL [68]	US	[8.40, 13.41]	RGB
	ARG [68]	Argentina	[3.90, 4.29]	RGB

2) Target Domain Datasets: The target domain dataset includes five widely used datasets (Houston [65], JAX [66], OMA [66], GeoNRW_Rural [67], and GeoNRW_Urban [67]) and two more challenging datasets (ARG [68], ATL [68]). Among them, the target domain formed by the widely used datasets is denoted as T.D.1, while the target domain formed by the more challenging datasets is denoted as T.D.2. The characteristics of these more challenging datasets include significant differences in the mean and standard deviation of height, as well as their origin from regions outside the United States and the European Union. For height estimation, all seven target domain datasets are used for evaluation; for semantic segmentation, only JAX and OMA are used, as these two datasets provide the land-cover category annotations necessary for evaluation. Table II introduces all the target domain datasets, detailing their geographic origin, statistical features of the height values (mean and standard deviation), and standardized imaging channel types.

We evaluate our method under three distinct architectural frameworks: ResNet101 + DeepLabV2 (CNN-based), MiT-B5 + DAFormer (Transformer-based), and DINOv2 + DPT (Transformer-based). All experiments were conducted on NVIDIA RTX 4090 GPUs. For data augmentation, we applied strong transformations such as ClassMix with a ratio of $|C|/2$, and standard augmentations including ColorJitter, GaussianBlur, and GaussNoise with probabilities of 0.8, 0.5, and 0.5, respectively. In the pseudolabel generation process, we set the land cover confidence threshold τ to 0.95 and the height map consistency thresholds η_1 and η_2 to 1.55 and 1.65, respectively. For optimization, we used the AdamW optimizer with an encoder learning rate of 1×10^{-6} and a decoder learning rate of 1×10^{-5}, with weight decay set to 5×10^{-4}. Training was conducted using a polynomial decay mode. The feature loss threshold ϵ was set to 0.6, and both the target loss and feature loss weighting coefficients were set to 1. The batch size for training was set to 2, with each batch containing one annotated synthetic image from SynRS3D and one unlabeled image from the target domain training set. The model was trained for 40 000 iterations. In addition, in DPCDO, the teacher model is updated using an exponential moving average of the student model parameters, with the update formula as follows:

$$\theta_t \leftarrow \alpha\theta_t + (1-\alpha)\,\theta_s \tag{20}$$

where θ_t represents the teacher model parameters, θ_s represents the student model parameters, and α is the EMA decay factor.

B. Evaluation Metric

To comprehensively evaluate and compare the performance of different models, we use intersection over union (IoU) and mean IoU (mIoU) for the semantic segmentation task. These metrics assess the overlap between predicted and ground truth labels at the pixel level and evaluate the overall performance across multiple categories. For height estimation, we employ mean absolute error (MAE) and root-mean-squared error (RMSE) to quantify the discrepancy between predicted and true height values.

Furthermore, to assess the precision of height prediction against ground truth, we introduce the threshold-based accuracy (δ) metric. The δ metric evaluates the ratio of predicted height to true height as follows:

$$\delta = \max\left(\frac{\hat{Y}}{Y}, \frac{Y}{\hat{Y}}\right) \tag{21}$$

where $\hat{Y}$ and Y represent the predicted and true height values, respectively.

To better evaluate the model's performance in predicting significant height targets (e.g., those greater than 1 meter), we use the height-aware $F1$ score (F_1^{HE} [30]). This metric focuses on assessing the prediction accuracy of meaningful height targets, such as buildings and trees, and avoids performance dominance by the large number of zero-value ground pixels. For example, in a remote sensing image where 90% of the pixels are ground (zero height), and 10% are buildings, a model that predicts all pixels as zero would have a very low overall MAE, but it would fail to detect any buildings. F_1^{HE} addresses this by ignoring ground pixels and concentrating on evaluating the model's ability to detect objects with height, thus preventing "false high accuracy." This metric, considering a predefined significance threshold T and relative error tolerance η, provides a more balanced evaluation by weighing both precision and recall. Its formula is defined as follows:

True positives (TPs)

$$TP = \sum\left(\left(\hat{Y} > T \wedge Y > T\right) \wedge \left(\delta < \eta\right)\right). \tag{22}$$

False positives (FPs)

$$FP = \sum\left(\hat{Y} > T \wedge Y \leq T\right). \tag{23}$$

False negatives (FNs)

$$FN = \sum\left(\hat{Y} \leq T \wedge Y > T\right). \tag{24}$$

The final score is computed using the standard F_1 formula

$$Precision = \frac{TP}{TP + FP} \tag{25}$$

$$Recall = \frac{TP}{TP + FN} \tag{26}$$

$$F_1^{HE} = 2 \times \frac{Precision \times Recall}{Precision + Recall}. \tag{27}$$

C. Comparison With Existing Methods

To maintain alignment with the RS3DAda method [30], we conduct quantitative evaluation on the "Building," "Tree," and "Ground" classes to ensure fair comparison. In addition, we

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TABLE III

SEMANTIC SEGMENTATION RESULTS ON JAX AND OMA DATASETS (MIOU ↑ PER CLASS AND OVERALL). THE BEST RESULTS FOR EACH BACKBONE ARE IN BOLD, AND THE OVERALL BEST ARE UNDERLINED. THE OVERALL MIOU IS REPORTED AS MEAN$_{\pm \text{std}}$

Backbone	Model	JAX[66]				OMA [66]				Avg. (J+O)			
		Ground	Tree	Building	mIoU	Ground	Tree	Building	mIoU	Ground	Tree	Building	mIoU
ResNet101+DeepLabV2	Source-only	68.93	54.18	57.98	60.36	75.54	43.01	43.76	54.10	72.24	48.60	50.87	$57.23_{\pm 1.637}$
	AdaptSeg [69]	73.21	55.70	58.76	62.56	75.54	45.25	50.62	57.14	74.38	50.48	54.69	$59.85_{\pm 1.022}$
	DADA [70]	70.11	57.92	62.70	63.58	77.06	**48.58**	52.56	59.40	73.59	**53.25**	57.63	$61.69_{\pm 0.955}$
	PIT [71]	72.12	55.88	63.02	63.67	77.56	47.27	56.13	60.32	74.84	51.58	59.58	$62.00_{\pm 0.871}$
	SAC [72]	73.03	57.66	63.10	64.60	77.84	47.25	56.71	60.60	75.44	52.46	59.91	$62.60_{\pm 0.863}$
	RS3DAda [30]	76.53	54.96	65.39	65.56	79.48	47.63	56.89	61.33	78.01	51.30	61.14	$63.45_{\pm 0.826}$
	Ours	**76.60**	**58.01**	**67.33**	**67.31**	**79.99**	48.12	**58.24**	**62.12**	**78.30**	53.07	**62.79**	$\mathbf{64.72}_{\pm 0.677}$
	Train-on-target	83.90	62.88	74.76	73.84	89.59	54.37	76.70	73.56	86.75	58.63	75.73	$73.70_{\pm 0.189}$
MiT-B5+DAFormer	HRDA [73]	74.66	54.21	64.35	64.41	77.89	46.99	56.77	60.55	76.27	50.60	60.56	$62.48_{\pm 1.077}$
	MIC [74]	75.12	54.26	64.65	64.58	77.81	47.24	56.69	60.58	76.45	50.75	60.67	$62.58_{\pm 0.936}$
	BUS [75]	76.37	54.55	65.31	65.41	78.26	47.44	57.03	60.91	77.32	50.99	61.17	$63.16_{\pm 0.761}$
	RS3DAda[30]	77.02	55.13	66.32	66.15	80.55	47.71	57.66	61.97	78.79	51.42	61.99	$64.06_{\pm 0.841}$
	SiamSeg [76]	77.64	56.23	66.55	66.81	81.26	**48.59**	57.93	62.59	79.45	**52.41**	62.24	$64.70_{\pm 0.736}$
	UniMAP [41]	77.95	57.50	66.80	67.41	81.65	47.00	59.21	62.62	79.80	52.25	63.01	$65.02_{\pm 0.611}$
	Ours	**79.01**	**59.10**	**68.62**	**68.91**	**84.14**	45.21	**61.63**	**63.66**	**81.57**	52.16	**65.12**	$\mathbf{66.28}_{\pm 0.607}$
	Train-on-target	85.43	65.24	77.91	76.19	90.33	57.13	79.66	75.70	87.88	61.19	78.79	$75.94_{\pm 0.143}$
(DINOv2+ViT-S)+DPT	DAFormer [77]	73.36	54.53	61.23	63.04	75.45	46.39	53.33	58.39	74.41	50.46	57.28	$60.82_{\pm 1.255}$
	RS3DAda [30]	76.89	55.05	65.93	65.96	79.92	**47.62**	57.32	61.62	78.41	**51.69**	61.56	$63.79_{\pm 0.901}$
	Ours	**77.59**	**58.88**	**67.12**	**67.86**	**82.66**	44.22	**59.85**	**62.24**	**80.13**	51.55	**63.49**	$\mathbf{65.05}_{\pm 0.874}$
	Train-on-target	84.21	63.70	75.92	74.61	89.64	55.67	77.66	74.32	86.53	59.69	76.79	$74.47_{\pm 0.215}$
(DINOv2+ViT-B)+DPT	DAFormer [77]	74.23	54.56	63.01	63.93	76.47	46.81	55.17	59.48	75.35	50.69	59.09	$61.70_{\pm 0.998}$
	RS3DAda [30]	77.02	55.10	66.20	66.10	80.53	**47.77**	57.61	61.97	78.76	51.46	61.91	$64.04_{\pm 0.895}$
	Ours	**78.86**	**59.79**	**68.23**	**68.96**	**84.11**	44.06	**61.34**	**63.17**	**81.49**	**51.93**	**64.79**	$\mathbf{66.07}_{\pm 0.641}$
	Train-on-target	85.12	64.18	77.36	75.56	90.37	56.13	79.32	75.27	87.75	60.16	78.34	$75.42_{\pm 0.172}$
(DINOv2+ViT-L)+DPT	DAFormer [77]	76.03	54.88	65.21	65.37	78.14	47.53	56.06	60.58	77.09	51.21	60.64	$62.97_{\pm 1.145}$
	RS3DAda [30]	77.38	55.26	66.69	66.44	80.91	**48.03**	57.87	62.27	79.16	51.65	62.28	$64.36_{\pm 0.824}$
	Ours	**79.62**	**60.17**	**68.89**	**69.56**	**85.09**	44.23	**63.53**	**64.28**	**82.36**	**52.20**	**66.21**	$\mathbf{66.92}_{\pm 0.569}$
	Train-on-target	85.67	65.86	78.69	76.74	90.64	58.15	80.50	76.43	88.16	62.01	79.60	$76.59_{\pm 0.110}$

TABLE IV

SEMANTIC SEGMENTATION RESULTS ON JAX AND OMA (MIOU ↑ PER CLASS AND OVERALL, AVG. (J + O)) UNDER SEMANTIC SEGMENTATION-ONLY TRAINING. THE BACKBONE OF THESE METHODS IS MIT-B5 + DAFORMER. THE OVERALL MIOU IS REPORTED AS MEAN$_{\pm \text{std}}$

Model	Ground	Tree	Building	mIoU
HRDA [73]	76.27	50.60	60.56	$62.48_{\pm 1.07}$
MIC [74]	76.45	50.75	60.67	$62.58_{\pm 0.93}$
RS3DAda [30]	77.06	50.63	61.23	$62.97_{\pm 0.81}$
BUS [75]	77.32	50.99	61.17	$63.16_{\pm 0.76}$
SiamSeg [76]	79.45	52.41	62.24	$64.70_{\pm 0.73}$
UniMAP [41]	79.80	52.25	63.01	$65.02_{\pm 0.61}$
Ours	**81.22**	**52.64**	**64.14**	$\mathbf{65.99}_{\pm 0.568}$

also include the "Water" class in the qualitative results to comprehensively demonstrate the model's performance across all cross-domain shared categories.

1) Quantitative Analysis: As illustrated in Table III, DPCDO consistently outperforms across all architectural frameworks, achieving the highest average mIoU of 66.92% on the JAX and OMA datasets. Notably, the method demonstrates significant improvements in the challenging building class, which is typically difficult to handle due to its irregular structures and complex geometry. This highlights DPCDO's effectiveness in addressing complex geometric structures, a critical challenge in remote sensing semantic segmentation. Furthermore, as more advanced architectures are employed, the performance gains become even more pronounced. For instance, DPCDO improves mIoU by 1.27 % with the ResNet101 + DeepLabV2 framework, while the improvement rises to 2.56% with the more modern (DINOv2 + ViT-L) + DPT architecture. This indicates a strong synergy between DPCDO and modern Transformer architectures, enabling it to leverage Transformers' ability to capture long-range dependencies and complex spatial structures, leading to more precise and efficient semantic understanding. Beyond the improvements in mean performance, DPCDO consistently achieves lower standard deviations across all backbone architectures. Under the (DINOv2 + ViT-L) + DPT framework, for instance, the standard deviation is reduced from ±0.824 to ±0.569, indicating more stable performance across independent runs. In addition, to examine whether the observed performance gains rely solely on joint training, we evaluate a setting where only semantic segmentation is trained (see Table IV). Given the different dependencies of each module on the joint task,

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TABLE V

HEIGHT ESTIMATION RESULTS ACROSS ALL TARGET DOMAIN DATASETS. THE BEST RESULTS FOR EACH BACKBONE ARE IN BOLD, AND THE OVERALL BEST ARE UNDERLINED. MAE-WHOLE IS REPORTED AS MEAN$_{\pm\text{std}}$

Backbone	Model	Avg. T.D.1 MAE ↓		Avg. T.D.2 MAE ↓		Avg. (T.D.1+T.D.2) MAE ↓		RMSE ↓		F_1^{HE} ↑		
		Whole	High	Whole	High	Whole	High	Whole	High	$\delta < 1.25$	$\delta < 1.25^2$	$\delta < 1.25^3$
ResNet101+DeepLabV2	Source-only	$2.391_{\pm 0.072}$	5.478	$5.056_{\pm 0.148}$	7.771	$3.153_{\pm 0.102}$	6.132	5.046	7.561	0.401	0.542	0.627
	DADA [70]	$2.030_{\pm 0.050}$	5.075	$4.582_{\pm 0.101}$	7.013	$2.759_{\pm 0.070}$	5.572	4.570	7.023	0.405	0.549	0.626
	RS3DAda [30]	$2.051_{\pm 0.019}$	4.488	$4.281_{\pm 0.038}$	6.866	$2.688_{\pm 0.026}$	5.255	4.459	6.911	0.423	0.566	0.638
	Ours	$\mathbf{2.017}_{\pm 0.014}$	**4.561**	$\mathbf{4.209}_{\pm 0.025}$	**6.262**	$\mathbf{2.643}_{\pm 0.018}$	**5.047**	**4.257**	**6.318**	**0.461**	**0.581**	**0.651**
	Train-on-target	$1.530_{\pm 0.007}$	3.760	$2.867_{\pm 0.015}$	5.062	$1.912_{\pm 0.010}$	4.132	3.625	5.228	0.522	0.664	0.732
(DINOv2+ViT-S)+DPT	RS3DAda [30]	$1.965_{\pm 0.021}$	4.859	$4.425_{\pm 0.044}$	6.506	$2.667_{\pm 0.030}$	5.532	4.411	6.890	0.428	0.573	0.642
	Ours	$\mathbf{1.944}_{\pm 0.014}$	**4.568**	$\mathbf{4.374}_{\pm 0.026}$	**6.117**	$\mathbf{2.637}_{\pm 0.018}$	**5.012**	**4.251**	**6.221**	**0.462**	**0.597**	**0.662**
	Train-on-target	$1.527_{\pm 0.009}$	3.753	$2.733_{\pm 0.018}$	4.541	$1.871_{\pm 0.012}$	3.976	3.293	5.227	0.587	0.696	0.739
(DINOv2+ViT-B)+DPT	RS3DAda [30]	$1.951_{\pm 0.016}$	4.570	$4.246_{\pm 0.033}$	6.121	$2.606_{\pm 0.022}$	5.013	4.163	6.279	0.440	0.581	0.651
	Ours	$\mathbf{1.868}_{\pm 0.011}$	**4.430**	$\mathbf{4.101}_{\pm 0.022}$	**6.120**	$\mathbf{2.506}_{\pm 0.015}$	**4.914**	**4.128**	**6.215**	**0.465**	**0.601**	**0.670**
	Train-on-target	$1.441_{\pm 0.007}$	3.495	$2.578_{\pm 0.013}$	4.230	$1.766_{\pm 0.008}$	3.705	3.106	4.911	0.629	0.729	0.764
(DINOv2+ViT-L)+DPT	RS3DAda [30]	$1.920_{\pm 0.013}$	4.415	$4.206_{\pm 0.027}$	6.260	$2.573_{\pm 0.018}$	4.943	4.155	6.209	0.466	0.608	**0.677**
	Ours	$\underline{\mathbf{1.844}}_{\pm 0.009}$	**4.413**	$\underline{\mathbf{3.902}}_{\pm 0.018}$	**5.987**	$\underline{\mathbf{2.423}}_{\pm 0.012}$	**4.863**	**4.005**	**6.144**	**0.482**	**0.616**	**0.677**
	Train-on-target	$1.419_{\pm 0.006}$	3.327	$2.542_{\pm 0.012}$	4.024	$1.740_{\pm 0.007}$	3.526	2.964	4.691	0.646	0.742	0.774

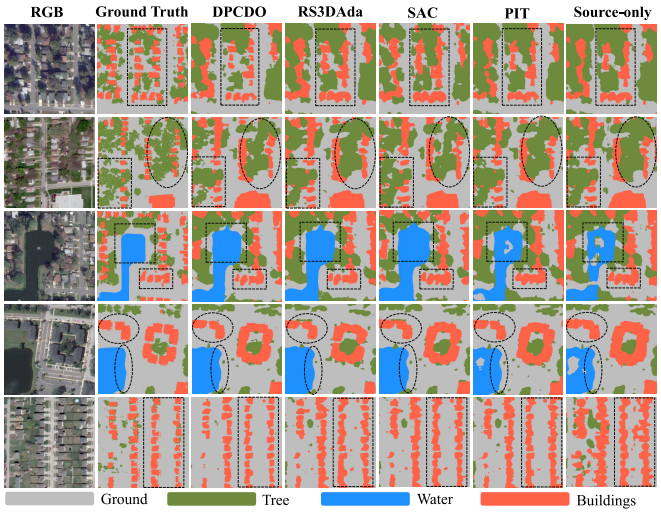


Fig. 4. Qualitative comparison of semantic segmentation on the target domain. (From left to right) Original RGB image, ground truth, and the predictions from the methods: DPCDO, RS3DAda, SAC, PIT, and Source-only.

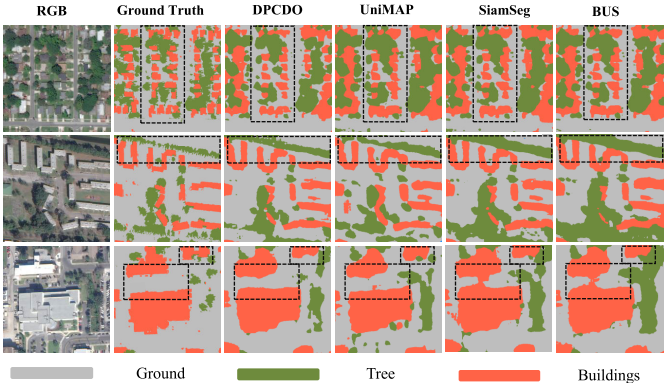


Fig. 5. Qualitative comparison of semantic segmentation on the target domain under semantic segmentation-only training. (From left to right) Original RGB image, ground truth, and the predictions from the methods: DPCDO, UniMAP, SiamSeg, and BUS.

our method retains only the DPPM module under this setting. The results show that even without sharing height-related features, our model is still able to maintain stable performance advantages.

As illustrated in Table V, DPCDO, based on the (DINOv2 + ViT-L) + DPT architecture, achieves new state-of-the-art performance in terms of MAE, reducing it from 2.573 to 2.423 m—a 5.8% improvement over the current best method. In addition, the improvements in F_1^{HE} are particularly notable: a gain of 1.6

2) Qualitative Analysis: Fig. 4 highlights DPCDO's advantages in boundary delineation and spatial accuracy. The method achieves two key improvements.

1) Enhanced edge definition—compared to the blurred and imprecise boundaries in SAC and RS3DAda, DPCDO generates sharper and clearer contours for buildings, trees, and water bodies, especially in areas with overlapping objects, effectively distinguishing adjacent land cover types.
2) Improved spatial fidelity—predicted regions align more accurately with ground truth labels, demonstrating stronger spatial alignment and geometric consistency, with building contours closely matching the ground truth. These improvements reflect the effectiveness of DPCDO's dynamic-coordinated learning mechanism in preserving fine-grained spatial details and achieving high-precision modeling. In addition, we present qualitative results obtained under the semantic segmentation-only setting (see Fig. 5), which further demonstrate that DPPM is able to preserve the aforementioned advantages in boundary delineation and spatial consistency even without joint training.

Fig. 6 demonstrates DPCDO's height prediction performance in complex urban scenes. The method achieves two key advantages.

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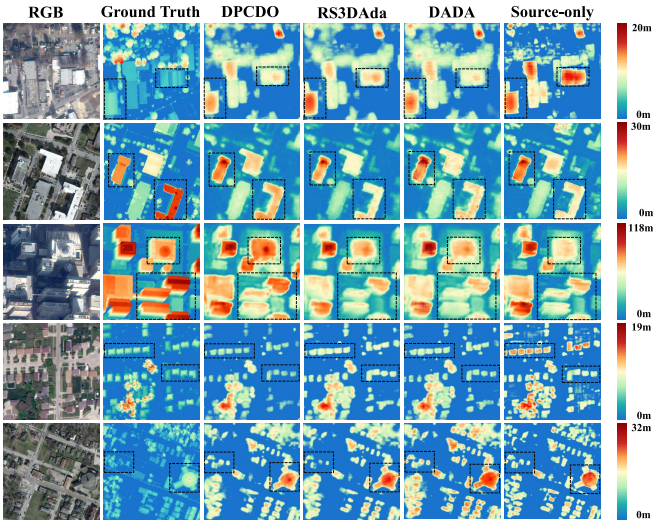


Fig. 6. Qualitative comparison of height estimation on the target domain. (From left to right) Original RGB image, ground truth, and the predictions from the methods: DPCDO, RS3DAda, DADA, and Source-only.

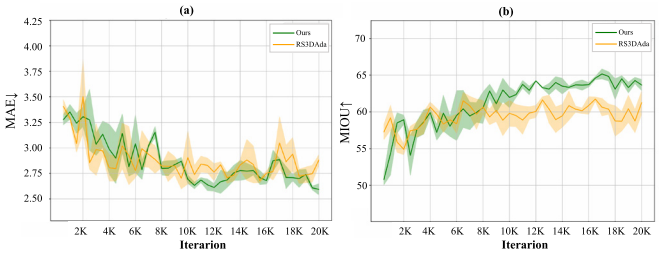


Fig. 7. Training dynamics and convergence analysis. Backbone: (DINOv2 + ViT-L) + DPT. (a) MAE trends in height estimation across target domains. (b) mIoU trends in semantic segmentation on JAX and OMA datasets. Shaded regions represent variance across multiple runs.

1) Clear height differentiation—DPCDO accurately maintains distinct height boundaries between different land cover types, particularly between buildings and surrounding terrain or roads, ensuring clear separation of height values.
2) Enhanced height consistency—compared to DADA and RS3DAda, the generated height maps exhibit colors that closely match the ground truth values, reducing overestimation and underestimation issues. These improvements demonstrate that DPCDO provides more accurate height predictions when dealing with complex urban scenes featuring varied terrain and building structures.

D. Training Dynamics and Convergence Analysis

Fig. 7 presents an analysis of the early-stage training dynamics. In the height estimation task, as shown in (a), DPCDO converges quickly after 10K iterations while consistently maintaining a low MAE. On the contrary, the RS3DAda method exhibits slower convergence with larger fluctuations in MAE, indicating that DPCDO demonstrates superior accuracy and convergence speed when generalizing to target domain features. For the semantic segmentation task, as illustrated in (b), DPCDO begins to outperform RS3DAda after 8K iterations, achieving a steady 5% improvement in mIoU by 11K iterations. This suggests that, as training progresses, DPCDO better adapts to different semantic categories, showing a significant performance gain. Overall, when handling complex cross-domain data, DPCDO exhibits better training stability, faster convergence, and smaller error margins, demonstrating clear advantages over existing methods.

E. Ablation Studies

Here, we analyze the contribution of each component by incrementally adding it to the baseline method under the joint training setting. In addition, to validate our decoupled optimization strategy, we compare DOMH against existing gradient conflict mitigation methods, including GradNorm, PCGrad, and CAGrad. Beyond this, we conduct isolated modeling experiments—training semantic segmentation and height estimation independently—to further validate module effectiveness in single-task scenarios. In the segmentation-only setting, only DPPM is retained, as MS2C is designed exclusively for height pseudolabel consistency, and DOMH requires dual-task decoders to function. Symmetrically, in the height estimation-only setting, only MS2C is evaluated, as DPPM's cross-task adaptation weights rely on segmentation-derived prototypes and lose their coupling benefit without joint training. Finally, we conduct ablation studies on the hyperparameter selection for the DPPM and MS2C modules.

1) Joint Training Component Ablation Studies: As illustrated in Table VI, for the semantic segmentation task, DPPM significantly improves performance on the "Building" class, yielding a + 3.25% increase on the JAX dataset. MS2C further enhances pseudolabel quality through multiscale consistency constraints, improving the average accuracy [Avg. (J + O)] of tree and building categories to 52.92% and 66.36%, respectively. The combination of DPPM, MS2C, and DOMH results in the best overall performance, achieving an mIoU of 66.92%, thus validating the effectiveness of the dynamic-coordinated learning mechanism. As seen in Table VII, for height estimation, DPPM reduces the MAE from 2.573 to 2.461 m, showing a significant performance improvement. MS2C offers additional gains through multiscale consistency. The full framework, incorporating DPPM, MS2C, and DOMH, achieves optimal results with an F_1^{HE} of 0.482 at $\delta < 1.25$, validating the effectiveness of the coordination strategy. In addition, to accurately evaluate the independent contribution of DOMH, we compare it against GradNorm, PCGrad, and CAGrad on top of the unified DPPM + MS2C baseline (Table VIII). DOMH achieves the best performance on both tasks simultaneously, yielding an mIoU of 66.92% and a MAE of 2.423 m. At the same time, it reduces the training time from 3.72 to 3.05 h per 10k iterations, matching the baseline overhead. These results indicate that the performance gains of DOMH are attributable to its alternating-update design itself: it decouples the gradient flows between the task-specific decoders while preserving the shared encoder representations, rather than relying on the joint training setting.

2) Isolated Modeling Component Ablation Studies: As illustrated in Tables IX and X, under segmentation-only training, DPPM improves mIoU from 63.58% to 66.34%, with particularly notable gains on the building class (+4.81%);

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TABLE VI

ABLATION ON MODULE EFFECTIVENESS FOR SEMANTIC SEGMENTATION (MIOU ↑ PER CLASS AND OVERALL). THE BACKBONE OF THESE METHODS IS (DINOV2 + VIT-L) + DPT

Model	JAX [66]				OMA [66]				Avg. (J+O)			
	Ground	Tree	Building	mIoU	Ground	Tree	Building	mIoU	Ground	Tree	Building	mIoU
Source-only	74.59	54.93	66.22	65.23	80.52	41.57	54.69	58.92	77.56	48.25	60.46	62.08
RS3DAda [30]	77.38	55.26	66.69	66.44	80.91	48.03	57.87	62.27	79.16	51.65	62.28	64.36
+DPPM	76.72	55.06	**69.94**	67.24	82.20	47.66	61.93	63.93	79.46	51.36	65.94	65.59
+MS2C	76.84	57.07	66.89	66.93	82.04	48.20	58.83	63.02	79.44	52.64	62.86	64.98
+DOMH	76.25	55.99	67.73	66.65	81.76	**48.73**	59.69	63.40	79.01	52.36	63.71	65.02
+DPPM+MS2C	77.35	57.47	69.28	68.03	83.47	48.36	63.43	**65.09**	80.41	**52.92**	**66.36**	66.56
+DPPM+MS2C+DOMH	**79.62**	**60.17**	68.89	**69.56**	**85.09**	44.23	**63.53**	64.28	**82.36**	52.20	66.21	**66.92**
Train-on-target	85.67	65.86	78.69	76.74	90.64	58.15	80.50	76.43	88.16	62.01	79.60	76.59

TABLE VII

ABLATION ON MODULE EFFECTIVENESS FOR HEIGHT ESTIMATION. THE BACKBONE OF THESE METHODS IS (DINOV2 + VIT-L) + DPT

Model	MAE ↓		RMSE ↓		$\mathbf{F_1^{HE}}$ ↑		
	Whole	High	Whole	High	$\delta < 1.25$	$\delta < 1.25^2$	$\delta < 1.25^3$
Source-only	3.007	6.111	4.977	7.561	0.411	0.556	0.633
RS3DAda [30]	2.573	4.943	4.155	6.209	0.466	0.608	0.677
+DPPM	2.461	5.038	4.087	6.338	0.457	0.593	0.658
+MS2C	2.519	4.940	4.123	6.256	0.463	0.602	0.671
+DOMH	2.530	**4.834**	4.106	6.148	0.457	0.597	0.666
+DPPM+MS2C	2.431	4.898	4.027	6.197	0.473	0.608	0.671
+DPPM+MS2C+DOMH	**2.423**	4.863	**4.005**	**6.144**	**0.482**	**0.616**	**0.677**
Train-on-target	1.740	3.526	2.964	4.691	0.646	0.742	0.774

TABLE VIII

COMPARISON OF GRADIENT CONFLICT MITIGATION STRATEGIES ON TOP OF THE DPPM + MS2C BASELINE. RESULTS ARE REPORTED IN TERMS OF MIOU (↑), MAE (↓), AND TRAINING TIME PER 10K ITERATIONS (↓). THE BACKBONE OF THESE METHODS IS (DINOV2 + VIT-L) + DPT

Method	DPPM	MS2C	GradNorm	PCGrad	CAGrad	DOMH	mIoU ↑	MAE ↓	Training Time ↓
M_1	✓	✓					66.56	2.431	**3.05**
M_2	✓	✓	✓				66.59	2.429	3.20
M_3	✓	✓		✓			66.58	2.427	3.66
M_4	✓	✓			✓		66.77	2.424	3.72
M_5	✓	✓				✓	**66.92**	**2.423**	**3.05**

TABLE IX

ABLATION ON MODULE EFFECTIVENESS UNDER SEMANTIC SEGMENTATION-ONLY TRAINING (MIOU ↑PER CLASS AND OVERALL, AVG. (J + O)). THE BACKBONE OF THESE METHODS IS (DINOV2 + VIT-L) +DPT

Model	Ground	Tree	Building	mIoU
RS3DAda [30]	79.10	50.43	61.22	63.58
+DPPM	**80.87**	**52.12**	**66.03**	**66.34**

under height estimation-only training, MS2C reduces MAE from 2.714 to 2.541 m and RMSE from 4.467 to 4.169 m on the whole evaluation set. These results demonstrate that both DPPM and MS2C retain strong standalone effectiveness independent of joint training, validating their applicability to single-task remote sensing domain adaptation scenarios.

3) Hyperparameter Ablation Studies: The performance of the DPPM module is influenced by the granularity of the prototypes, which is determined by the patch size. We examined the impact of different patch sizes on both semantic segmentation and height estimation tasks. As illustrated in Table XI, the results demonstrate that the 4×4 patch size yields the best performance across both tasks, effectively balancing prototype granularity and representation robustness. Smaller patches generate an excessive number of prototypes, leading to oversegmentation and noise, while larger patches may fail to adequately capture the complexity of the scene. On the contrary, the 4×4 patch size achieves an optimal balance between these factors, facilitating effective cross-task performance.

The multistream MS2C module employs a dual-threshold strategy to filter reliable pseudolabels at different decoder layers. Since the predictions from the final decoder layer

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TABLE X

ABLATION ON MODULE EFFECTIVENESS UNDER HEIGHT ESTIMATION-ONLY TRAINING. THE BACKBONE OF THESE METHODS IS (DINOV2 + VIT-L) + DPT

Model	MAE ↓		RMSE ↓	
	Whole	High	Whole	High
RS3DAda [30]	2.714	5.412	4.467	6.533
+MS2C	**2.541**	**5.013**	**4.169**	**6.289**

TABLE XI

IMPACT OF PATCH SIZE IN DPPM ON PERFORMANCE METRICS

Patch Size	Number of Prototypes	mIoU ↑	MAE ↓
2×2	196	66.06	2.491
4×4	49	66.92	2.423
7×7	16	64.99	2.475

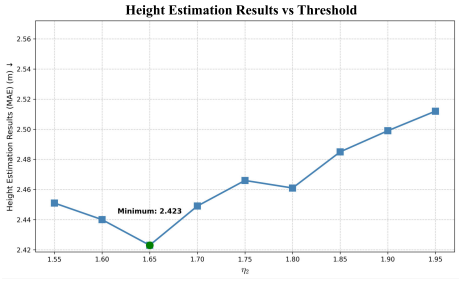


Fig. 8. Impact of different η_2 values on the height estimation task performance while maintaining $\eta_1 = 1.55$.

are better aligned with the height regression task and exhibit higher stability, a stricter threshold ($\eta_1 = 1.55$) is applied to ensure high-confidence pseudolabels. On the contrary, the auxiliary layer produces intermediate predictions that are more sensitive to data augmentations. To avoid overfiltering valid supervision signals, a relatively looser threshold is adopted for the auxiliary layer ($\eta_2 = 1.65$), allowing it to act as a filter only when obvious inconsistencies occur, thereby maintaining sufficient reliable supervision while preserving pseudolabel quality. To further validate the auxiliary threshold selection, we conduct a systematic analysis of η_2. As shown in Fig. 8, the MS2C module achieves optimal performance when $\eta_2 = 1.65$, yielding a MAE of 2.423 m. Higher thresholds (e.g., $\eta_2 = 1.70$) introduce noisy pseudolabels, whereas lower thresholds (e.g., $\eta_2 = 1.55$) remove valuable training signals, indicating that $\eta_2 = 1.65$ provides a good balance between pseudolabel reliability and supervision coverage.

F. Computational Cost Analysis

As illustrated in the Table XII, we evaluated the model's parameter count and training time per 10 000 iterations, where the parameter count reflects the computational resource requirements and the training time indicates the computational overhead. Due to the use of the DINOv2 (ViT-L) + DPT architecture, even the source-only method, without domain adaptation strategies, already has a large parameter count. With domain adaptation strategies, the RS3DAda method has a parameter count of 366.54M, while our proposed DPPM and DOMH modules introduce no learnable parameters. MS2C adds only a small number of parameters (to introduce auxiliary outputs, we added a convolutional layer to process the third-layer features from the decoder and generate additional outputs). While DPPM and MS2C slightly increase the computational time, the DOMH module introduces virtually no additional time overhead. This allows our method to enhance performance while maintaining low computational costs, making it suitable for deployment in resource-constrained environments.

TABLE XII

COMPUTATIONAL COST ANALYSIS OF OUR METHODS

Model	Params (M)	Training Time (h/10k iter.)
Source-only	335.32	2.25
RS3DAda [30]	366.54 (↑31.22)	2.80 (↑0.55)
+DPPM	366.54 (↑0.00)	2.92 (↑0.12)
+DPPM+MS2C	366.61 (↑0.07)	3.05 (↑0.13)
+DPPM+MS2C+DOMH	366.61 (↑0.00)	3.05 (↑0.00)

VI. CONCLUSION

We propose DPCDO, a novel dynamic-coordinated learning approach for multitask domain adaptation that systematically addresses domain shifts and task conflicts in remote sensing imagery. DPCDO can precisely capture domain discrepancies and effectively avoid task conflicts, achieving an adaptive balance between domain-invariant representation learning and cross-task collaboration. Experiments on multiple remote sensing benchmark datasets demonstrate that DPCDO outperforms existing state-of-the-art methods in both semantic segmentation and monocular height estimation, delivering consistently superior cross-domain performance. Furthermore, the method is easy to integrate with excellent domain adaptability and can be extended to other dense prediction tasks, providing an efficient and general solution for remote sensing multitask learning. To ensure reproducibility and facilitate future research, we have submitted the complete implementation code.

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